1.

a) (2.5 pts) For the wave described by $\mathbf{E}_1 = 4\sin(\omega t - \beta z)\hat{x}\,\frac{\text{V}}{\text{m}}$, the corresponding phasor is

$$\tilde{\mathbf{E}} = -\hat{x}4je^{-j\beta z}$$

b) (2.5 pts) For the wave described by $\mathbf{E}_2 = 5\cos(\omega t - \beta z)\hat{x} - 5\cos(\omega t - \beta z)\hat{y}\,\frac{\text{V}}{\text{m}}$,the corresponding phasor is

$$\tilde{\mathbf{E}}_2 = \hat{x}5e^{-j\beta z} - \hat{y}5e^{-j\beta z} = 5e^{-j\beta z}\,(\hat{x} - \hat{y})\,\frac{\text{V}}{\text{m}}.$$

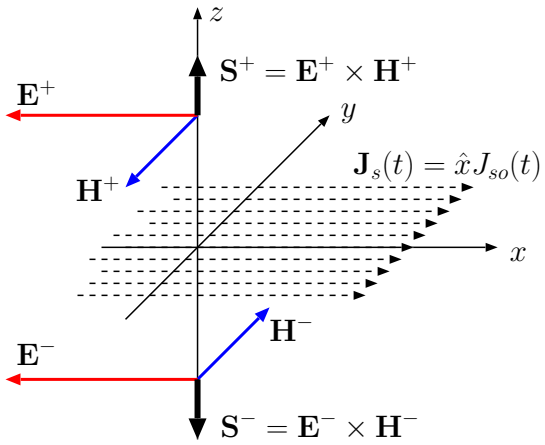
c) (2.5 pts) For the wave described by $\mathbf{H}_3 = 2\cos(\omega t + \beta z + \frac{\pi}{3})\hat{x} - 2\cos(\omega t + \beta z - \frac{\pi}{6})\hat{y}\,\frac{\text{A}}{\text{m}}$,the corresponding phasor is

$$\tilde{\mathbf{H}}_3 = \hat{x}2e^{j\beta z}e^{j\frac{\pi}{3}} - \hat{y}2e^{j\beta z}e^{-j(\frac{\pi}{2}-\frac{\pi}{3})} = 2e^{j(\beta z+\frac{\pi}{3})}\,(\hat{x} + j\hat{y})\,\frac{A}{\text{m}}.$$

d) (2.5 pts) For the wave described by $\mathbf{H}_4 = 4\cos(\omega t + \beta x)\hat{z} + 4\sin(\omega t + \beta x)\hat{y}\,\frac{\text{A}}{\text{m}}$,the corresponding phasor is

$$\tilde{\mathbf{H}}_4 = \hat{z}4e^{j\beta x} - \hat{y}4je^{j\beta x} = 4e^{j\beta x}\,(\hat{z} - j\hat{y})\,\frac{\text{A}}{\text{m}}.$$

2. The pulse of sheet current $\mathbf{J}_s(t) = \hat{x}J_{so}\cos(\omega t)\,\frac{\text{A}}{\text{m}}$ will produce magnetic and electric fields as shown in the following figure.



a) (3 pts) The average Poynting vector is $\langle S\rangle_{z>0} = \frac{1}{2}E_oH_o\,\hat{z}$ for the wave propagating in $+z$ direction while it is $\langle S\rangle_{z<0} = -\frac{1}{2}E_oH_o\,\hat{z}$ for the wave propagating in $-z$ direction. Since the average power density provided by the current sheet is $8\,\frac{\text{W}}{\text{m}^2}$, we can write that

$$\left|\langle S\rangle_{z>0}\right| + \left|\langle S\rangle_{z<0}\right| = E_oH_o = \frac{E_o^2}{\eta_o} = 8\,\frac{\text{W}}{\text{m}^2}.$$

Knowing that $\eta_o = 120\pi\,\Omega$, we get

$$E_o = \sqrt{8\eta_o} \approx 54.92\,\frac{\text{V}}{\text{m}},$$

and also

$$H_o = \frac{E_o}{\eta_o} \approx 0.146\,\frac{\text{A}}{\text{m}}.$$

b) (3 pts) The amount of instantaneous electromagnetic density power injected by the current surface is

$$-\mathbf{J}_s(t)\cdot\mathbf{E}(0,t) = J_{so}E_o\cos^2(\omega t),$$

from which we obtain the average power density

$$\langle -\mathbf{J}_s(t)\cdot\mathbf{E}(0,t)\rangle = \frac{1}{2}J_{so}E_o = 8\,\frac{\text{W}}{\text{m}^2},$$

which implies that

$$J_{so} = \frac{16}{E_o} \approx 0.291\,\frac{\text{A}}{\text{m}}.$$

c) (4 pts) The phasors $\tilde{\mathbf{E}}$ of part (a) is given by

$$\tilde{\mathbf{E}} = -E_o e^{\mp j\beta z}\,\hat{x},$$

whereas the phasor $\tilde{\mathbf{H}}$ is written as

$$\mathbf{H} = \pm H_o e^{\mp j\beta z}\,\hat{y}.$$

Then for $z > 0$, the phasors will be

$$\tilde{\mathbf{E}} = -54.92e^{-j\beta z}\,\hat{x}\,\frac{\text{V}}{\text{m}} \qquad \text{and} \qquad \tilde{\mathbf{H}} = -0.146e^{-j\beta z}\,\hat{y}\,\frac{\text{A}}{\text{m}}.$$

While for $z < 0$, the phasors will be

$$\tilde{\mathbf{E}} = -54.92e^{j\beta z}\,\hat{x}\,\frac{\text{V}}{\text{m}} \qquad \text{and} \qquad \tilde{\mathbf{H}} = 0.146e^{j\beta z}\,\hat{y}\,\frac{\text{A}}{\text{m}}.$$

3. **Damped waves - imperfect dielectric :** The electric field phasor of a plane wave is given by $\tilde{\mathbf{E}} = 4e^{-(0.0001+j0.2)z}\hat{x}\,\frac{\text{V}}{\text{m}}$.

a) (1 pt) We know that the propagation velocity, also called phase velocity, is $v_p = \frac{\omega}{\beta}$. Therefore, the wave frequency $f = \frac{\beta}{2\pi}v_p = \frac{0.2}{2\pi}\times 10^8 = \frac{1}{\pi}\times 10^7 = 3.183\,\text{MHz}$.

b) (1 pt) Wavelength is $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{0.2} = 10\pi = 31.416\,\text{m}$.

c) (2 pts) We know that the propagation velocity is $v_p = \frac{1}{\sqrt{\mu\epsilon}}$. Therefore, the dielectric constant is given by $\epsilon_r = \frac{\epsilon}{\epsilon_o} = \frac{1}{\mu\epsilon_o v_p^2}$ which yields

$$\epsilon_r = \frac{1}{(16\pi\times 10^{-7})\times\left(\frac{1}{36\pi}\times 10^{-9}\right)\times 10^{16}} = 2.25.$$

d) (2 pts) The attenuation constant is given in the problem as $\alpha = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$ which is the attenuation constant α for the case of an imperfect dielectric. Then, the conductivity is given by $\sigma = \frac{2\alpha}{\sqrt{\frac{\mu}{\epsilon}}} = \frac{2\alpha}{\eta_o}\sqrt{\frac{\epsilon_r}{\mu_r}}$ which yields

$$\sigma = \frac{2\times 0.0001}{120\pi}\times\sqrt{\frac{9/4}{4}} = 3.98\times 10^{-7}\,\frac{\text{S}}{\text{m}}.$$

e) (2 pts) The distance at which the peak amplitude of $\tilde{\mathbf{E}}$ is attenuated by $1/e$ is defined as the *skin depth* and is given by

$$z = \delta = \frac{1}{\alpha} = 10^4\,\text{m}.$$

f) (2 pts) The magnetic field phasor is given by

$$\tilde{\mathbf{H}} = \frac{4}{\eta} e^{-(0.0001+j0.2)z}\hat{y}\,\frac{\text{A}}{\text{m}},$$

where $\frac{1}{\eta} = \frac{1}{\eta_o}\sqrt{\frac{\epsilon_r}{\mu_r}}$. Then, the final expression of $\tilde{\mathbf{H}}$ is

$$\tilde{\mathbf{H}} = \frac{1}{40\pi} e^{-(0.0001+j0.2)z}\hat{y}\,\frac{\text{A}}{\text{m}}.$$

4. **Propagation in sea water :**

a) (2 pts) Since $\frac{\sigma}{\omega\epsilon} = \frac{4}{2\pi\times 10^2\cdot\frac{9}{4\pi}\times 10^{-9}} = \frac{8}{9}\times 10^7 \gg 1$, the medium behaves as a good conductors, i.e. $\alpha = \beta = \sim \sqrt{\pi f \mu\sigma}$.

b) (3 pts) With the approximation in (a), we find

$$\alpha = \beta \approx \sqrt{\pi 10^2 \cdot 4\pi\times 10^{-7}\cdot 4} \approx 0.0397\,\text{m}^{-1}.$$

Therefore the propagation constant is given by

$$\gamma = \alpha + j\beta = 0.0397 + j0.0397\,\text{m}^{-1}.$$

The wavelength is

$$\lambda = \frac{2\pi}{\beta} = 158.26\,\text{m},$$

whereas the penetration depth in the media with the intrinsic impedance

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}}e^{j\pi/4} = \sqrt{\frac{2\pi\times 10^2\times 4\pi\times 10^{-7}}{4}}\left(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right) = \pi\times 10^{-5/2}\,(1+j)\ \Omega,$$

is calculated as

$$\delta = \frac{1}{\alpha} \approx 25.19\,\text{m}.$$

c) (3 pts) Since $\frac{\sigma}{\omega\epsilon} = \frac{4}{2\pi\times 10^9\cdot\frac{9}{4\pi}\times 10^{-9}} = \frac{8}{9} \approx 1$, we will not be able to use the approximations listed in the first table of Lecture 24. Then, The propagation constant is given by $\gamma = \sqrt{j\omega\mu\,(\sigma + j\omega\epsilon)}$ which yields

$$\begin{aligned}\gamma &= \sqrt{(j2\pi\times 10^9\times 4\pi\times 10^{-7})(4 + j2\pi\times 10^9\times\frac{9}{4\pi}\times 10^{-9})}\\ &= 20\pi\sqrt{j8-9} = 77.48 + j203.8\,\text{m}^{-1}.\end{aligned}$$

Since $\alpha = 77.48\,\text{m}^{-1}$ and $\beta = 203.8\,\text{m}^{-1}$, the penetration depth and the wavelength are

$$\delta = \frac{1}{\alpha} = 1.291\times 10^{-2}\,\text{m} \quad \text{and} \quad \lambda = \frac{2\pi}{\beta} = 3.083\times 10^{-2}\,\text{m}.$$

Finally, the intrinsic impedance is

$$\begin{aligned}\eta &= \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} \\ &= \sqrt{\frac{j2\pi \times 10^9 \times 4\pi \times 10^{-7}}{4 + j2\pi \times 10^9 \times \frac{9}{4\pi} \times 10^{-9}}} \\ &= 40\pi\sqrt{\frac{j}{8 + j9}} \\ &= 33.85 + j12.87\,\Omega.\end{aligned}$$

d) (2 pts) From our calculations in (b) and (c), the penetration depths for 10^2 Hz and 10^9 Hz are 25.19 m and 1.291×10^{-2} m, respectively. Therefore, transmission signal from the submarine should use the lower frequency, i.e., 10^2 Hz to reduce the attenuation of the signal.

5. **Bonus problem (easy) -Transmission line characteristics:**

a) (5 pts) The geometrical factor of the RG-58 coax cable is given by

$$\text{GF} = \frac{2\pi}{\ln\left(\frac{0.056}{0.016}\right)} = 5.015,$$

from which we obtain

$$\begin{aligned}\mathcal{L} &= \frac{\mu_o}{\text{GF}} = \frac{4\pi \times 10^{-7}}{2\pi/\ln(3.5)} = 2.506 \times 10^{-7}\,\frac{\text{H}}{\text{m}} = 250.6\,\frac{\text{nH}}{\text{m}}, \\ \mathcal{C} &= 2.25\epsilon_o\text{GF} \approx \frac{2.25 \times 10^{-9}}{36\pi}\frac{2\pi}{\ln(3.5)} = 99.8\,\frac{\text{pF}}{\text{m}}, \\ Z_o &= \frac{1}{\text{GF}}\sqrt{\frac{\mu_o}{2.25\epsilon_o}} \approx \frac{\ln(3.5)}{2\pi}\frac{120\pi}{\sqrt{2.25}} = 50.11\,\Omega, \\ v &= \frac{1}{\sqrt{2.25\epsilon_o\mu_o}} = \frac{2}{3}c \approx 2 \times 10^8\,\frac{\text{m}}{\text{s}}.\end{aligned}$$

b) (5 pts) For twin-lead transmission lines, the geometrical factor is given by

$$\text{GF} = \frac{\pi}{\cosh^{-1}\left(\frac{D}{2a}\right)}.$$

Since

$$Z_o = \sqrt{\frac{\mathcal{L}}{\mathcal{C}}} = \frac{1}{\text{GF}}\sqrt{\frac{\mu}{\epsilon}} = \frac{\cosh^{-1}\left(\frac{D}{2a}\right)}{\pi}\sqrt{\frac{\mu}{\epsilon}},$$

we can can find that

$$D = 2a\,\cosh\left(Z_o\pi\sqrt{\frac{\epsilon}{\mu}}\right).$$

Assuming $\epsilon = \epsilon_o$, $\mu = \mu_o$, and $a = 1\,\text{mm} = 10^{-3}\,\text{m}$, let us calculate D as follows:

i. For $Z_o = 75\,\Omega$,

$$D = 2 \times 10^{-3}\,\cosh\left(\frac{75\pi}{120\pi}\right) \approx 2.40 \times 10^{-3}\,\text{m} = 2.4\,\text{mm}.$$

ii. For $Z_o = 450\,\Omega$,

$$D = 2 \times 10^{-3} \cosh\left(\frac{450\pi}{120\pi}\right) = 42.54 \times 10^{-3}\,\text{m} = 42.54\,\text{mm}.$$